**PART B** Please complete in BLOCK CAPITALS

**Student Number** ______________ **Name** ______________________________ **Signature** ______________________________

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**PART A** Please complete clearly

**Exam Number**
as shown on your university card

THE UNIVERSITY of EDINBURGH

# Financial Mathematics
# MATH10003

Monday 28 April 2025
14:30–16:30

**Number of questions: 3**
**Total number of marks: 100**

## IMPORTANT PLEASE READ CAREFULLY

### Before the examination

1. Put your university card face up on the desk.
2. **Complete PART A and PART B above**. By completing PART B you are accepting the University Regulations on student conduct in an examination (see back cover).
3. Complete the attendance slip and leave it on the desk.
4. This is a closed-book examination. No notes, printed matter or books are allowed.
5. A scientific calculator is permitted in this examination. It must not be a graphic calculator. It must not be able to communicate with any other device.

### During the examination

1. Write clearly, in ink, in the space provided after each question. If you need more space then please use the extra pages at the end of the examination script or ask an invigilator for additional paper.
2. **Attempt all questions**.
3. If you have rough work to do, simply include it within your overall answer – put brackets at the start and end of it if you want to highlight that it is rough work.

### At the end of the examination

1. This examination script must not be removed from the examination venue.
2. There are extra pages for working at the end of this examination script. If used, you should clearly label your working with the question to which it relates.
3. Additional paper and graph paper, if used, should be attached to the back of this examination script. Write your examination number on the top of each additional sheet.

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Formula Sheet:

1. Discrete compounding: $B_n = B_0(1+r)^n$.
2. Continuous compounding: $B_t = B_0 e^{rt}$.
3. Present Value of a bond under discrete compounding interest rate:

$$PV_B = C\frac{1}{r}\Big(1 - \frac{1}{(1+r)^T}\Big) + \frac{F}{(1+r)^T},$$

   $C$: coupon, $T$: time to maturity, $F$: face.
4. Payoff of European Call with strike $K$ and maturity $T$: $(S_T - K)^+$.
5. Payoff of European Put with strike $K$ and maturity $T$: $(K - S_T)^+$.
6. Fair delivery price of a forward contract: $K(t,T) = S_t e^{r(T-t)}$.
7. Law of One Price: In an arbitrage-free market, two portfolios (or assets or cashflows) with identical future values must have identical present values.
8. Put-Call Parity: $c_t + Ke^{-r(T-t)} = p_t + S_t$, where $c_t$ is premium of the call at time $t$, and $p_t$ is premium of the put at time $t$.
9. Value process in multi-period model: $V_n^h = x_n B_n + y_n S_n = x_n(1+r)^n + y_n S_n$.
10. A portfolio strategy $h_n = (x_n, y_n)$ is said to be a self-financing strategy if

$$x_n B_n + y_n S_n = x_{n+1} B_n + y_{n+1} S_n$$

   for all $n = 1, \ldots, N-1$.
11. The multi-period model is free of arbitrage if and only if $d < 1 + r < u$ (discrete comp.).
12. Risk-neutral probabilities in multi-period models (discrete comp.):

$$S_n = \frac{1}{1+r}\mathbb{E}^{\mathbb{Q}}\left[S_{n+1}\right], \quad q_u = \frac{(1+r)-d}{u-d}, \quad q_d = \frac{u-(1+r)}{u-d}.$$

13. Replicating portfolio for the claim whose contract function is $\Phi$ in 1-period model (discrete comp.):

$$x = \frac{1}{1+r}\frac{u\Phi(d) - d\Phi(u)}{u-d}, \quad y = \frac{1}{S_0}\frac{\Phi(u)-\Phi(d)}{u-d}.$$

14. Risk-neutral pricing for European type claim $X = \Phi(S_N)$ in $N$-period market(discrete comp.):

$$\Pi_0(X) = \left(\frac{1}{1+r}\right)^N \sum_{k=0}^{N} \binom{N}{k} q_u^k q_d^{N-k} \Phi\left(S_0 u^k d^{N-k}\right).$$

15. Risk-neutral pricing for American type claim $X = \Phi(S_N)$ in $N$-period market(discrete comp.):

$$\Pi_n(X)\big|_{S_n=s} = \max\left\{\Phi(s), \frac{1}{1+r}\Big(q_u \Pi_{n+1}(X)\big|_{S_{n+1}=us} + q_d \Pi_{n+1}(X)\big|_{S_{n+1}=ds}\Big)\right\}.$$

16. A stochastic process $W = (W_t)_{t\geq 0}$ is called a Brownian motion if it satisfies:

(a) $\mathbb{P}(W_0 = 0) = 1$.

(b) For all $0 \le s < t$, the random variable $W_t - W_s \sim N(0, t-s)$.

(c) $W$ has independent increments, i.e. for any $r < s \le t < u$ then $W_u - W_t$ and $W_s - W_r$ are independent random variables.

(d) Almost all sample paths are continuous.

17. Normal distribution $N(\mu, \sigma^2)$ has probability density function $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2}$.

18. Let $(X, Y)$ be bivariate normal and $\mathbb{E}[X] = \mathbb{E}[Y] = 0$, then $X$ and $Y$ are independent iff $\mathbb{E}[XY] = 0$.

19. Itô isometry:

$$\mathbb{E}\left[\left(\int_0^T X_s \mathrm{d}Ws\right)^2\right] = \mathbb{E}\left[\int_0^T |X_s|^2 \mathrm{d}s\right], \quad \mathbb{E}\left[\left(\int_0^T X_s \mathrm{d}W_s\right)\left(\int_0^T Y_s \mathrm{d}W_s\right)\right] = \mathbb{E}\left[\int_0^T X_s Y_s \mathrm{d}s\right].$$

20. Properties of Itô integrals:

(a) $\int_S^T g_t \mathrm{d}W_t = \int_S^U g_t \mathrm{d}W_t + \int_U^T g_t \mathrm{d}W_t$,

(b) $\int_0^T (\alpha g_t + f_t)\, \mathrm{d}W_t = \alpha \int_0^T g_t \mathrm{d}W_t + \int_0^T f_t \mathrm{d}W_t$,

(c) $\mathbb{E}\left[\int_0^t g_s \mathrm{d}W_s\right] = 0$.

21. Itô's formula: for $Y_t := f(t, X_t)$,

$$\begin{aligned} \mathrm{d}Y_t &= \frac{\partial f}{\partial t}(t, X_t)\mathrm{d}t + \frac{\partial f}{\partial x}(t, X_t)\mathrm{d}X_t + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}(t, X_t)\langle \mathrm{d}X_t, \mathrm{d}X_t\rangle \\ &= \left[\frac{\partial f}{\partial t}(t, X_t) + U_t \frac{\partial f}{\partial x}(t, X_t) + V_t^2 \frac{1}{2}\frac{\partial^2 f}{\partial x^2}(t, X_t)\right]\mathrm{d}t + \left[V_t \frac{\partial f}{\partial x}(t, X_t)\right]\mathrm{d}W_t, \end{aligned}$$

where $\mathrm{d}X_t = U_t \mathrm{d}t + V_t \mathrm{d}W_t$.

22. $\langle \mathrm{d}t, \mathrm{d}t\rangle = \langle \mathrm{d}t, \mathrm{d}W_t\rangle = \langle \mathrm{d}W_t, \mathrm{d}t\rangle = 0$, $\langle \mathrm{d}W_t, \mathrm{d}W_t\rangle = \mathrm{d}t$.

23. Geometric Brownian motion:

$$\mathrm{d}S_t = \mu S_t \mathrm{d}t + \sigma S_t \mathrm{d}W_t, \quad S_t = S_0 \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W_t\right), \quad \mathbb{E}(S_t) = S_0 e^{\mu t}$$

24. Black-Scholes PDE:

$$\frac{\partial f}{\partial t}(t, x) + rx\frac{\partial f}{\partial x}(t, x) + \frac{1}{2}\sigma^2 x^2 \frac{\partial^2 f}{\partial x^2}(t, x) - rf(t, x) = 0$$

with the terminal condition

$$f(T, x) = \Phi(x),$$

where $\Phi$ is the payoff function of the option, $\sigma > 0$ is the volatility and $r$ is the interest rate.

25. Feynman-Kac formula: Assume that the function $f : [0, T] \times \mathbb{R} \to \mathbb{R}$ is a solution to the following terminal value problem PDE

$$\frac{\partial f}{\partial t}(t, x) + \mu(t, x)\frac{\partial f}{\partial x}(t, x) + \frac{1}{2}\sigma^2(t, x)\frac{\partial^2 f}{\partial x^2}(t, x) - rf(t, x) = 0, \tag{1}$$

$$f(T, x) = \Phi(x). \tag{2}$$

For any $(t,x) \in [0,T] \times \mathbb{R}$, define the stochastic process $(X_s)_{s\in[t,T]}$ as the solution to the SDE

$$\mathrm{d}X_s = \mu(s,X_s)\mathrm{d}s + \sigma(s,X_s)\mathrm{d}W_s, \quad \forall\, s \in [t,T], \qquad X_t = x.$$

Assume that the stochastic process $\left(e^{-rs}\sigma(s,X_s)\frac{\partial f}{\partial x}(s,X_s)\right)_{s\in[t,T]} \in L^2([0,T]\times\mathbb{R})$.

Then the solution $f$ of (1)-(2) can be expressed as

$$f(t,x) = e^{-r(T-t)}\mathbb{E}[\Phi(X_T)|X_t = x] \qquad \forall\,(t,x) \in [0,T]\times\mathbb{R}.$$

1. Consider a two-period binomial model, in which a non-dividend-paying stock $S = (S_t)_{t=0,1,2}$ has its current price of $S_0 = £120$. Over the next two 4-month periods, the stock price is expected to either increase by 10% or decrease by 6%. Let us assume that the risk-free rate of interest is 6% per annum (continuous compounding).

   (a) Show that the market is free of arbitrage. Then find the risk-neutral probabilities. Please round your solution to 1 decimal place. [6 marks]

If you have used additional space for working then please tick here: ☐

(b) Suppose there is an 8-month European call option written on $S$ with strike price $K = £124$.

i. What is the arbitrage-free price of this option at $t = 0$? Please round your answer to 3 decimal places. [7 marks]

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If you have used additional space for working then please tick here: ☐

ii. Suppose Amy bought this option at $t = 0$, then decided to sell this contract after 4 months. How much should Amy sell it for after 4 months (i.e. at $t = 1$) so that there is no arbitrage opportunity? Please round your solution to 3 decimal places. [6 marks]

If you have used additional space for working then please tick here: ☐

iii. Suppose Katie short sells this call option immediately at $t = 0$. To hedge her short position in this option, Katie adopts a delta-hedging strategy by taking a long position in $S$. Assuming she can hold a fractional number of stock units, how many units of stock must she buy to achieve a perfect hedge? Round your answer to 3 decimal places. By showing the portfolio is riskless (round the answer to two decimal places), verify that it results in a perfect hedge. [6 marks]

If you have used additional space for working then please tick here: ☐

(c) Find the arbitrage-free price of an 8-month American put option written on $S$ with strike price $K = £124$. Please round your answer to 3 decimal places. Is an early exercise optimal? [13 marks]

[Total for this question: 38 marks]

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If you have used additional space for working then please tick here: ☐

2. Let $T > 0$. Assume a Black-Scholes market, where the evolution of stock price $(S_t)_{t\in[0,T]}$ is given by a geometric Brownian motion with drift $r$ and volatility $\sigma$, and the bank account pays risk-free interest continuously at annual rate $r$, so that

$$S_t = s_0 \exp\left(\left(r - \frac{1}{2}\sigma^2\right) t + \sigma W_t\right),$$

where $(W_t)_{t\geq 0}$ is a standard Brownian motion, and $s_0 > 0$ is a real number.

Let $n \in \mathbb{N}$ be a positive integer. Consider the *power call* option $\Phi$ whose payoff function is given by

$$\Phi(x) = (x^n - K)^+$$

and the *power put* option $\Phi$ whose payoff function is given by

$$\Phi(x) = (K - x^n)^+.$$

(a) Verify that $(S_t)_{t\geq 0}$ satisfies the stochastic differential equation

$$\begin{aligned} \mathrm{d}S_t &= rS_t\mathrm{d}t + \sigma S_t \mathrm{d}W_t, \\ S_0 &= s_0. \end{aligned}$$

Then find the stochastic differential equation satisfied by the $n$-th power of the stock price process, $(S_t^n)_{t\geq 0}$, which is also a geometric Brownian motion. [9 marks]

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If you have used additional space for working then please tick here: ☐

(b) Find the Black-Scholes price (or arbitrage-free price) $\Pi_0(\mathrm{Call}(T,K))$ of the power call option with strike $K$, maturity $T$. Write down your answer using the cumulative distribution function (CDF) $N(x)$ of the standard normal distribution $N(0,1)$, $d_-(n)$ and $d_+(n)$, where

$$d_-(n) = \frac{n(r-\frac{\sigma^2}{2})T + \ln\left(\frac{s_0^n}{K}\right)}{n\sigma\sqrt{T}},$$

and

$$d_+(n) = d_-(n) + n\sigma\sqrt{T} = \frac{n(r-\frac{\sigma^2}{2}+n\sigma^2)T + \ln\left(\frac{s_0^n}{K}\right)}{n\sigma\sqrt{T}}.$$

**Hint**: For $K>0$, find $\mathbb{P}(S_T^n > K)$ and $\mathbb{E}[S_T^n 1_{\{S_T^n>K\}}]$. [15 marks]

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If you have used additional space for working then please tick here: ☐

(c) Find the Black-Scholes price (or arbitrage-free price) $\Pi_0(\mathrm{Put}(T, K))$ of the power put option with strike $K$, maturity $T$. [8 marks]

If you have used additional space for working then please tick here: ☐

(d) Using the explicit expressions for $\Pi_0(\mathrm{Call}(T,K))$ and $\Pi_0(\mathrm{Put}(T,K))$ found above, write down the Put-Call parity relation for power options. [4 marks]

[Total for this question: 36 marks]

If you have used additional space for working then please tick here: ☐

3. (a) In the *Vasicek interest rate model*, the interest rate is a stochastic process $(R_t)_{t\geq 0}$ which evolves according to the following stochastic differential equation

$$\mathrm{d}R_t = (b - aR_t)\mathrm{d}t + \sigma \mathrm{d}W_t, \quad R_0 = r,$$

where $(W_t)_{t\geq 0}$ is a standard Brownian motion, and $r, a, b, \sigma > 0$ are positive real numbers.

i. By considering $X_t = (R_t - \frac{b}{a}(1 - e^{-at}))e^{at}$, show that $(R_t)_{t\geq 0}$ is given by

$$R_t = re^{-at} + \frac{b}{a}(1 - e^{-at}) + \sigma e^{-at} \int_0^t e^{as} \mathrm{d}W_s, \quad \forall t \geq 0.$$

[6 marks]

If you have used additional space for working then please tick here: ☐

ii. From the above representation, you may assume that for every $t > 0$, $R_t$ has a normal distribution. Find its expectation $\mathbb{E}[R_t]$ and variance $\mathrm{Var}(R_t)$. [8 marks]

If you have used additional space for working then please tick here: ☐

(b) Let $T, \sigma, r, \theta > 0$ be fixed constants. Use the Feynman-Kac representation formula to solve the following equation subject to a terminal condition:

$$\frac{\partial u}{\partial t}(t,x) + \frac{1}{2}\sigma^2 \frac{\partial^2 u}{\partial x^2}(t,x) = ru(t,x), \quad 0 \le t \le T,$$
$$u(T,x) = e^{\theta x}.$$

[12 marks]

[Total for this question: 26 marks]

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If you have used additional space for working then please tick here:

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**The University of Edinburgh Exam Hall Regulations**

1. An examination attendance sheet is laid on the desk for each student to complete upon arrival. These are collected by an invigilator after thirty minutes have elapsed from the start of the examination. Students are not allowed to enter the examination hall more than thirty minutes after the start of the examination.
2. Students arriving after the start of the examination are required to complete a “Late arrival form” which requires them to sign a statement that they understand that they are not entitled to any additional time. Students are not allowed to leave the examination hall less than thirty minutes after the commencement of the examination or within the last fifteen minutes of the examination.
3. Personal belongings e.g. coats, jackets, any form of headwear (with the exception of headwear worn for religious reasons) electronic equipment, bags, books, papers, briefcases and cases must be left at the front/back or sides of the examination room. No coats or jackets are permitted on the back of chairs. It is a breach of the Code of Student Conduct for a student to have in their possession in the examination any material relevant to the work being examined unless this has been authorised by the examiners.
4. Students must take their seats within the block of desks allocated to them and must not communicate with other students either by word or sign, nor let their papers be seen by any other student.
5. Students are prohibited from deliberately doing anything that might distract other students. Students wishing to attract the attention of an invigilator shall do so without causing a disturbance. Any student who causes a disturbance in an examination room may be required to leave the room, and may be reported by the invigilator.
6. An announcement will be made to students that they may start the examination. Students must stop writing immediately when the end of the examination is announced. Students who continue to write after the end of the examination will be reported to their School.
7. Answers should be written in ink (unless otherwise instructed) in the script book provided. Rough work, if any, should be completed within the script book and subsequently crossed out. Script books must be left in the examination hall.
8. Dictionaries, reference books, computers, calculators, electronic devices or any other material are NOT permitted. The only exceptions are:-
    1. If such is specified in a student’s Learning Profile as assessed by the Student Disability Service, or
    2. For all students taking courses where the provision of a dictionary, reference books, computers, calculators, electronic devices or any other material is an integral part of the assessment process. Such exceptional arrangements requires authorisation by the examiners for use by all students and is recorded in the exam’s instructions.
9. The use of mobile devices/personal electronic equipment is not permitted. Mobile devices must be switched off during an examination. These should be placed in your bag and should not be on your person. Mobile devices are those which store/display data or connect to the internet, such as a mobile telephone, smart watches, smart glasses or any other communications equipment.
10. It is a breach of the Code of Student Conduct for any student knowingly
    - to make use of unfair means in any University examination
    - to assist a student to make use of such unfair means
    - to do anything prejudicial to the good conduct of the examination, or
    - to impersonate another student or allow another student to impersonate them
11. Students will be required to display their University card on the desk throughout all written degree examinations and certain other examinations. If a card is not produced, the student will be required to make alternative arrangements to all his her identity to be verified before examinations is marked.
12. Smoking (including the use of e-cigarettes) and eating is not permitted inside the examination hall. Only students who can provide a letter from Student Disability Services, that grants them permission to eat, will be allowed to do so. Students who have received this letter of permission are required to show the letter to the invigilator.
13. If an invigilator suspects a student of cheating, they shall impound any prohibited material and shall inform the Examinations department as soon as possible. A report will also be sent to the relevant School.
14. Cheating is an extremely serious offence, and any student found by the University to have cheated or attempted to cheat in an examination may be deemed to have failed that examination or the entire diet of examinations, or be subject to such penalty as the University considers appropriate.